

Student Name _____ Student Class _____



2020

HSC Open Book Test Term 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Board approved calculators may be used.
- Write using black or blue pen.
- All necessary working should be shown in all questions.
- Write your student number at the top of every page of your answer sheets.
- You may bring in 1 page A4 paper with anything written or typed on it.

Total marks – 61

Attempt all questions

Start each question on a new sheet of paper

A Formula Sheet is provided

This paper MUST NOT be removed from the examination room

Section A Multiple Choice (5 Marks)

QUESTION 1

We can express $\sin x$ and $\cos x$ in terms of $\tan \frac{x}{2}$, for all values of x except.....

(A) $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

(B) $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$

(C) $x = \pi, 3\pi, 5\pi$

(D) $x = 2\pi, 6\pi, 8\pi$

QUESTION 2

What is the value of k if $\int_0^k \frac{dx}{9+x^2} = \frac{\pi}{9}$

(A) $\sqrt{3}$

(B) 3

(C) $\frac{1}{\sqrt{3}}$

(D) $3\sqrt{3}$

QUESTION 3

What is the coefficient of x^3 in the expansion $(1 - 3x)^7$

(A) -945

(B) -35

(C) 35

(D) 945

QUESTION 4

An advertisement claims that '8 out of 10 prefer Winky Chocolate Bars'. If the advertisement's claim is accurate and a sample of six people are interviewed, what is the probability that at least five people prefer Winky Chocolate Bars?

- (A) $1 - (0.8)^4$
- (B) $2(0.8)^5$
- (C) $5(0.2)^5$
- (D) $(0.8)^5(0.2) + (0.8)^6$

QUESTION 5

What is the derivative of $y = \cos^{-1}(\frac{1}{x})$ with respect to x ?

- (A) $\frac{-1}{\sqrt{x^2 - 1}}$
- (B) $\frac{-1}{x\sqrt{x^2 - 1}}$
- (C) $\frac{1}{\sqrt{x^2 - 1}}$
- (D) $\frac{1}{x\sqrt{x^2 - 1}}$

Section B Answer on the lined paper provided**Marks****QUESTION 6 (8 Marks) Start a new page**

(a) Evaluate $\int_0^1 \frac{1}{\sqrt{4-2x^2}} dx$ leaving your answer in exact form.

3

(b) Show that $\int_0^{\frac{\pi}{16}} \cos^2 4x dx$ is $\frac{1}{16} + \frac{\pi}{32}$

2

(c) Solve the equation $\sin 2x = \tan x$ for $0 \leq x \leq \pi$

3**QUESTION 7 (8 Marks) Start a new page**

(a) Extensive research has shown that 1 person out of every 4 is allergic to a particular grass seed. A group of university students volunteer to try out a new treatment

(i) If 20 students are tested, what is the expected number of allergic people in the group?

1

(ii) Find how large a sample would need to be for the probability of it containing at least one allergic person, to be greater than 99.9%?

3

(iii) What assumptions have you made in your answer?

1

(b) Evaluate $\int_1^{49} \frac{1}{4(x+\sqrt{x})} dx$ using the substitution $u^2 = x$, $u > 0$

3

Give the answer in exact form.

QUESTION 8 (8 Marks) Start a new page

(a) Use the fact that for a restricted domain, $y = \sin^{-1} x$ is equivalent to

2

$x = \sin y$ to show that $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

(b) A spherical weather balloon is being inflated from empty using a container of helium, so that its volume is increasing at a constant rate of $0.5 \text{ m}^3/\text{s}$. (Assume the balloon maintains a spherical shape throughout inflation.)

(i) Show that the radius at a time t is given by $r = \sqrt[3]{\frac{3t}{8\pi}}$

2

(ii) Show that the rate of increase of its surface area after 8 seconds is $\sqrt[3]{\frac{\pi}{3}} \text{ m}^2/\text{s}$.

3

(iii) If the maximum safe surface area before there is a risk that the balloon will burst is 200 m^2 , what is the maximum time that the inflation should be allowed to proceed?

1

QUESTION 9 (8 Marks) Start a new page

(a) $X_1, X_2, \dots, X_n$ are independent $B(1, p)$ random variables.

Let $Y = X_1 + X_2 + \dots + X_n$ and $\hat{p} = \frac{Y}{n}$

(i) Given $\text{var}(Y) = np(1-p)$ show $\text{var}(\hat{p}) = \frac{p(1-p)}{n}$

2

(ii) Show $0 \leq \text{var}(\hat{p}) \leq \frac{1}{4n}$

2

(b) Let X be a Bernoulli random variable with parameter p , where $0 < p < 1$

(i) Explain why p cannot take the values $p = 0$ or $p = 1$.

1

(ii) How many distinct values can X take in n Bernoulli trials?

1

(iii) A biased coin is tossed where $P(\text{Head}) = \frac{1}{5}$. If you obtain Heads you win a prize of \$1.

Could this situation be modelled as a Bernoulli random variable? Explain

2

QUESTION 10 (8 Marks) Start a new page

- (a) Use the substitution $t = \tan \frac{\theta}{2}$ to prove the identity

$$\frac{1}{2}(3\cos\theta + 4\sin\theta + 5) = \left(\sin\frac{\theta}{2} + 2\cos\frac{\theta}{2}\right)^2 \quad 3$$

- (b) Show that $\binom{n+2}{3} - \binom{n}{3} = n^2$ for all integers n , where $n \geq 3$. 2

- (c) Suppose that the true, unknown proportion p of voters who will vote for a third-party candidate in the next election is 9%. What is the probability that a poll of 820 voters will find a sample proportion $\hat{p}$ that differs from the true proportion by more than 3%? You may use the normal distribution on the Reference sheet to approximate your answer to the nearest 0.1% 3

QUESTION 11 (8 Marks) Start a new page

- (a) The blades of a wind turbine are turning at a steady rate. The height, h metres, of the tip of one of the blades above the ground at time, t seconds, is given by the formula

$$h = 36\sin(1.5t) - 15\cos(1.5t) + 65$$

- (i) Express $36\sin(1.5t) - 15\cos(1.5t)$ in the form $k\sin(1.5t - a)$, where $k > 0$ and $0 < a < 2\pi$, 3

- (ii) Hence find the two values of t for which the tip of this blade is at a height of 100 metres above the ground during the first turn. 3

- (b) For the expansion $(1 - 2x - 4x^2)\left(1 - \frac{3}{x}\right)^9$, find the coefficient of the term in x^{-5} 2

QUESTION 12 (8 Marks) Start a new page

In the expansion of $(0.15 + 0.85)^{50}$, the terms involving $(0.15)^r$ and $(0.15)^{r+1}$ are denoted by T_r and T_{r+1}

(i) Show that $\frac{T_r}{T_{r+1}} = \frac{17(r+1)}{3(50-r)}$ 4

The probability that Paul's train to work is late on any day is 0.15, independently of other days. The number of days on which Paul's train is late during a 50-day period is modelled by the random variable X

(ii) Find the values of r for which $P(X = r) \leq P(X = r + 1)$ 3

(iii) Hence find the most likely number of days on which the train will be late during the 50-day period. 1

END OF EXAMINATION



Student Number _____

Question 6

$$\begin{aligned} (a) \int_0^1 \frac{1}{\sqrt{4-2x^2}} dx &= \frac{1}{\sqrt{2}} \int_0^1 \frac{1}{\sqrt{2-x^2}} dx \quad \checkmark \quad \sqrt{4-2x^2} = \sqrt{2(2-x^2)} \\ &= \frac{1}{\sqrt{2}} \left[\sin^{-1} \left(\frac{x}{\sqrt{2}} \right) \right]_0^1 \\ &= \frac{1}{\sqrt{2}} \left(\left[\sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \right] - \sin^{-1} \left(\frac{0}{\sqrt{2}} \right) \right) \checkmark \\ &= \frac{1}{\sqrt{2}} \left[\frac{\pi}{4} - 0 \right] \\ &= \frac{\pi}{4\sqrt{2}} \checkmark \\ &= \frac{\sqrt{2}\pi}{8} \end{aligned}$$

$$\begin{aligned} (b) \int_0^{\frac{\pi}{16}} \cos^2 4x dx &= \frac{1}{2} \int_0^{\frac{\pi}{16}} \cos 8x + 1 dx \quad \checkmark \\ &= \frac{1}{2} \left[\frac{\sin 8x}{8} + x \right]_0^{\frac{\pi}{16}} \\ &= \frac{1}{2} \left[\frac{\sin \frac{\pi}{2}}{8} + \frac{\pi}{16} - \left(\frac{\sin 0}{8} + 0 \right) \right] \checkmark \\ &= \frac{1}{2} \left(\frac{1}{8} + \frac{\pi}{16} \right) \\ &= \frac{1}{16} + \frac{\pi}{32} \end{aligned}$$

$$(c) \sin 2x = \tan x \cdot 0 \leq x \leq \pi$$

$$2 \sin x \cos x = \frac{\sin x}{\cos x}$$

$$2 \sin x \cos^2 x = \sin x \quad \checkmark$$

$$\sin x (2 \cos^2 x - 1) = 0$$

$$\therefore \sin x = 0 \quad \text{OR} \quad 2 \cos^2 x - 1 = 0 \quad \checkmark$$

$$x = 0, \pi$$

$$\cos^2 x = \frac{1}{2}$$

$$\cos x = \pm \sqrt{\frac{1}{2}}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\therefore x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi \quad \checkmark$$

MATHEMATICS Extension 1: Question 7

Suggested Solutions

Marks

Marker's Comments

(a) Expected number of allergic people

$$(i) E(X) = 0.25 \times 20 = 5$$

$\therefore 5$ allergic people in group ✓

(ii) $P(\text{at least 1 allergic person}) = 1 - (0.75)^n$

$$1 - (0.75)^n > 0.999$$

$$(0.75)^n < 0.001$$

$$n \ln(0.75) < \ln 0.001$$

$$\therefore n > \frac{\ln 0.001}{\ln 0.75}$$

$$n > 24.61$$

$\therefore 25$ people are required.

(iii) accepted reasons: assumptions

- Random
- Independent
- Unbiased
- sample represents population
- probability of being allergic remains constant for all people
- Uni students have the same chance of being allergic as all other people

1

3

✓

1

1 mark for recognising $1 - 0.75^n > 0.999$
1 mark solving for n - looking at steps
1 mark for getting $n = 24.61$ and rounding it up to get 25

MATHEMATICS Extension 1: Question 7

Suggested Solutions

Marks

Marker's Comments

$$7(b) \int_1^{49} \frac{1}{4(x+\sqrt{x})} dx$$

let $u^2 = x$
 $\frac{dx}{du} = 2u$
 $\therefore dx = 2u \cdot du$

when $x = 1$
 $u = 1$
 when $x = 49$
 $u = 7$

$$\int_1^7 \frac{1}{4(u^2+u)} \cdot 2u \cdot du$$

$$= \frac{1}{2} \int_1^7 \frac{u}{u^2+u} \cdot du$$

$$= \frac{1}{2} \int_1^7 \frac{1}{u+1} \cdot du$$

$$= \frac{1}{2} [\ln(u+1)]_1^7$$

$$= \frac{1}{2} [\ln 8 - \ln 2]$$

$$= \frac{1}{2} [\ln 4]$$

$$= \ln 2$$

3

✓ 1 mark for substituting
 $* dx = 2u \cdot du$
 $* x = u^2$
 $* \sqrt{x} = u$

✓ 1 mark for integrating with correct integral.
 ie $\frac{1}{2} \int_1^7 \frac{1}{u+1} \cdot du$

Note this is not equal to
 $\frac{1}{2} \int_1^7 \frac{1}{u} + 1$

✓ 1 substituting with correct limits $u=1$ and $u=7$
 accepted
 $\frac{1}{2} [\ln 8 - \ln 2]$

Question 8b)

Monday, 20 April 2020 12:18 pm

$$(i) \frac{dV}{dt} = 0.5$$

$$\therefore V = 0.5t + C \quad \text{when } t=0, V=0$$

$$\therefore C=0$$

$$V = 0.5t$$

$$V = \frac{4}{3}\pi r^3$$

$$0.5t = \frac{4}{3}\pi r^3$$

$$t = \frac{8}{3}\pi r^3$$

$$r^3 = \frac{3t}{8\pi}$$

$$\therefore r = \sqrt[3]{\frac{3t}{8\pi}}$$

$$\textcircled{1} V = 0.5t \text{ after showing } C=0$$

$$\textcircled{1} \text{ expression for } t$$

ii) Method 1: Simplest

$$S = 4\pi r^2$$

$$\frac{dS}{dr} = 8\pi r \quad \frac{dt}{dr} = 8\pi r^2$$

$$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt} = 8\pi r \times \frac{1}{8\pi r^2}$$

$$= \frac{1}{r}$$

$$= \sqrt[3]{\frac{8\pi}{3t}} \quad r = \sqrt[3]{\frac{3t}{8\pi}}$$

$$\text{when } t=8$$

$$\frac{dS}{dt} = \sqrt[3]{\frac{8\pi}{3 \times 8}} = \sqrt[3]{\frac{\pi}{3}}$$

Method 2:

$$S = 4\pi r^2, \quad r = \sqrt[3]{\frac{3t}{8\pi}} = \sqrt[3]{\frac{3}{8\pi}} \times t^{1/3}$$

$$\frac{dS}{dr} = 8\pi r$$

$$\frac{dr}{dt} = \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3} t^{-2/3} = \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3\sqrt[3]{t^2}}$$

$$\frac{dS}{dt} = \frac{dS}{dr} \times \frac{dr}{dt} = 8\pi r \times \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3\sqrt[3]{t^2}} = 8\pi \times \sqrt[3]{\frac{3t}{8\pi}} \times \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3\sqrt[3]{t^2}}$$

$$\text{when } t=8$$

$$\begin{aligned} \frac{dS}{dt} &= 8\pi \times \sqrt[3]{\frac{3}{\pi}} \times \sqrt[3]{\frac{3}{8\pi}} \times \frac{1}{3} \sqrt[3]{\frac{1}{8^2}} \\ &= \frac{8\pi}{3} \times \sqrt[3]{\frac{9}{8^3 \pi^2}} \\ &= \frac{9\sqrt[3]{8^3 \times 9\pi^3}}{3^3 8^3 \pi^2} \\ &= \sqrt[3]{\frac{\pi}{3}} \end{aligned}$$

Marks:

$$\textcircled{1} \text{ for } \frac{dS}{dt} = \frac{1}{r}$$

$$\textcircled{1} \frac{dS}{dt} \text{ in terms of } t$$

$$\textcircled{1} \text{ substitution of } t=8 \text{ and simplification.}$$

$$(iii) \quad 4\pi r^2 = 200$$

$$r^2 = \frac{200}{4\pi}$$

$$= \frac{50}{\pi}$$

$$r = \sqrt{\frac{50}{\pi}}, \quad r > 0$$

$$\therefore \sqrt{\frac{50}{\pi}} = \sqrt[3]{\frac{3t}{8\pi}}$$

$$t = \left(\sqrt{\frac{50}{\pi}}\right)^3 \times \frac{8\pi}{3} = \frac{8 \times 50 \sqrt{50}}{3 \sqrt{\pi}} = \frac{400 \sqrt{50}}{3 \sqrt{\pi}}$$

$$= 531.9230405 \text{ s}$$

$$\therefore \text{max. time} = 531.9 \text{ s}$$

$$\text{or } 531.9 \text{ s}$$

$$\text{or } 8 \text{ min } 51.9 \text{ s}$$

$\left. \begin{array}{l} \textcircled{1} \text{ no penalty for} \\ \text{subsequent error} \\ \text{of rounding up} \\ \text{to } 532 \text{ s which} \\ \text{is too long.} \end{array} \right\}$

Question 8a)

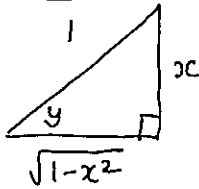
Monday, 20 April 2020

12:07 pm

a) $y = \sin^{-1} x$

$x = \sin y$

$\frac{dx}{dy} = \cos y$

Method 1:

$\cos y = \sqrt{1-x^2}$

$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

Method 2:

$\cos^2 y + \sin^2 y = 1$

$\cos^2 y = 1 - \sin^2 y$

$\cos y = \sqrt{1 - \sin^2 y}$

since for $y = \sin^{-1} x$, $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ and $\cos x > 0$ for this domain

$\therefore \frac{dx}{dy} = \sqrt{1 - \sin^2 y}$

$\frac{dy}{dx} = \frac{1}{\sqrt{1 - \sin^2 y}}$

$= \frac{1}{\sqrt{1-x^2}}$

① for finding $\frac{dx}{dy} = \cos y$ Method 1:① for using triangle to show $\cos y = \sqrt{1-x^2}$

and $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

note that $\frac{dy}{dx} \neq \frac{dx}{dy}$

① for using Pythagorean identity to

show $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$

Suggested Solutions	Marks	Marker's Comments
a) i) $x_1, x_2, \dots, x_n, B(1, p)$ $y = x_1 + x_2 + \dots + x_n, \hat{p} = \frac{y}{n}$ } given. $Var(Y) = np(1-p)$ Show $var(\hat{p}) = \frac{p(1-p)}{n}$ we know $Var(Y) = npq, q = (1-p)$ <u>M1</u> $\begin{aligned} var(\hat{p}) &= var\left(\frac{Y}{n}\right) \\ &= E\left(\left(\frac{Y}{n}\right)^2\right) - E\left(\frac{Y}{n}\right)^2 \\ &= \frac{1}{n^2} E(Y^2) - \frac{1}{n^2} [E(Y)]^2 \\ &= \frac{1}{n^2} [E(Y^2) - (E(Y))^2] \\ &= \frac{1}{n^2} Var(Y) \\ &= \frac{1}{n^2} (np(1-p)) \text{ from above} \\ &= \frac{p(1-p)}{n} \text{ shown.} \end{aligned}$ <u>M2</u> Given $Var(Y) = np(1-p)$ $Var(\hat{p}) = var\left(\frac{Y}{n}\right)$ $= \frac{1}{n^2} Var(Y) \rightarrow$ (since variance is the square of std dev) $= \frac{1}{n^2} (np(1-p))$ $= \frac{p(1-p)}{n}$ (shown) example from maths in Fom 12 (Ext 1) pg 692.	1 1	

Suggested Solutions

Marks

Marker's Comments

aii) Show $0 \leq \text{var}(\hat{p}) \leq \frac{1}{4n}$

$$y = p(1-p)$$

$$\frac{dy}{dp} = 1 - 2p$$

$$\text{when } \frac{dy}{dp} = 0$$

$$p = \frac{1}{2}$$

p	0	$\frac{1}{2}$	1
$\frac{dy}{dp}$	1	0	-1
shape/sign	/	-	\

$$y_{\text{max}} \text{ at } p = \frac{1}{2}$$

$$= \frac{1}{2} \left(1 - \frac{1}{2}\right)$$

$$= \frac{1}{4}$$

- greatest value of $p(1-p)$ occurs when $p = \frac{1}{2}$; $p(1-p) = \frac{1}{4}$

- Smallest value of $p(1-p)$ occurs

when $p = 0$ or 1 ; $p(1-p) = 0$

- smallest value of $\text{var}(\hat{p}) = \frac{0}{n} = 0$

- greatest value of $\text{var}(\hat{p}) = \frac{1}{4} = \frac{1}{4n}$

$$\therefore 0 \leq \text{var}(\hat{p}) \leq \frac{1}{4n}$$

b)i) - If $p = 0$ or $p = 1$, there is a 100% chance that the trial will be a failure or success:

- A Bernoulli trial is an experiment that has 2 outcomes, either failure or success.

- If there is a 100% chance of either occurring, that means that there is only one outcome.

$\therefore$ The experiment will not be a Bernoulli trial becoz p cannot be 0 or 1

① - mark to show understanding of stationary points and concave down

① - mark for stating max tp at $p = \frac{1}{2}$ and y_{max} at $\frac{1}{4}$

or have shown a graphical interpretation.

② For why $\text{var}(\hat{p}) \geq 0$.

① - mark.

Have to mention something along the line why/how Bernoulli trial happens.

Suggested Solutions	Marks Awarded	Marker's Comments
b) ii) How many distinct values can X take in 'n' Bernoulli trials? The answer is <u>2</u> or <u>0 or 1</u>	1	Many students interpreted it incorrectly. The question said $X = \text{Bernoulli random variable}$ <u>NOT</u> Binomial random variable - This was the issue For Bernoulli random variable - there are only 2 distinct values X can take. i.e. 2 or 0 or 1
b) iii) $P(H) = \frac{1}{5}$ if heads win \$1 Is this Bernoulli random variable, (BRV) ie from my comment, BRV has 2 outcomes; if $P(H) = \frac{1}{5}$ $P(\bar{H}) = \frac{4}{5}$ win lose. (1) (0) as per definition. Therefore the answer is yes and the second mark is awarded if student explains this correctly.	1 1	1 1

Q10.

a) Let $t = \tan \frac{\theta}{2}$ (Method 1)

$$\therefore \frac{1}{2} (3 \cos \theta + 4 \sin \theta + 5)$$

$$= \frac{1}{2} \left(3 \times \frac{1-t^2}{1+t^2} + 4 \times \frac{2t}{1+t^2} + 5 \right)$$

$$= \frac{1}{2} \left(\frac{3-3t^2}{1+t^2} + \frac{8t}{1+t^2} + \frac{5+5t^2}{1+t^2} \right)$$

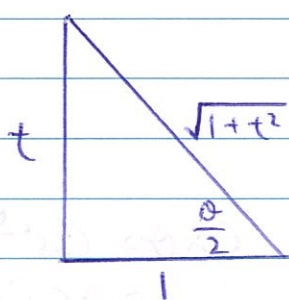
$$= \frac{1}{2} \left(\frac{3-3t^2+8t+5+5t^2}{1+t^2} \right)$$

$$= \frac{1}{2} \left(\frac{2t^2+8t+8}{1+t^2} \right)$$

$$= \frac{t^2+4t+4}{1+t^2}$$

$$= \frac{(t+2)^2}{1+t^2}$$

① Arriving at this expression from LHS



$$\sin \frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}}$$

$$\cos \frac{\theta}{2} = \frac{1}{\sqrt{1+t^2}}$$

① This step is a must (including the triangle) as students are not allowed to quote $\sin \frac{\theta}{2}$ and $\cos \frac{\theta}{2}$ in a "prove" question

$$\therefore \text{RHS} = \left(\frac{t}{\sqrt{1+t^2}} + 2 \times \frac{1}{\sqrt{1+t^2}} \right)^2$$

$$= \left(\frac{t+2}{\sqrt{1+t^2}} \right)^2$$

$$= \frac{(t+2)^2}{1+t^2}$$

$$= \text{LHS}$$

① Arriving at this expression from RHS.

or After arriving LHS = $\frac{(t+2)^2}{1+t^2}$ like in method 1

(Method 2)

$$\text{LHS} = \frac{(t+2)^2}{1+t^2} \quad \text{--- (1)}$$

$$= \frac{\left(\tan \frac{\theta}{2} + 2\right)^2}{1 + \tan^2 \frac{\theta}{2}}$$

$$= \frac{\left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} + 2\right)^2}{\sec^2 \frac{\theta}{2}}$$

Getting rid of t or $\tan \frac{\theta}{2}$ in the expression

$$= \cos^2 \frac{\theta}{2} \left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} + 2 \right)^2$$

$$= \left[\cos \frac{\theta}{2} \left(\frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} + 2 \right) \right]^2$$

$$= \left(\sin \frac{\theta}{2} + 2 \cos \frac{\theta}{2} \right)^2$$

Arriving to RHS. (1)

What NOT to do

Intention of Qn

$$\text{RHS} = \left(\sin \frac{\theta}{2} + 2 \cos \frac{\theta}{2} \right)^2$$

$$= \sin^2 \frac{\theta}{2} + 4 \sin \frac{\theta}{2} \cos \frac{\theta}{2} + 4 \cos^2 \frac{\theta}{2}$$

$$= 1 + 3 \cos^2 \frac{\theta}{2} + 4 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$= 1 + \frac{3 \cos \theta + 3}{2} + 2 \sin \theta \quad *$$

$$\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$$

$$= 2 \cos^2 \frac{\theta}{2} - 1$$

$$\therefore \cos^2 \frac{\theta}{2} = \frac{\cos \theta + 1}{2}$$

$$\text{Sub } \cos \theta = \frac{1-t^2}{1+t^2} \text{ and } \sin \theta = \frac{2t}{1+t^2}$$

$$\therefore \text{RHS} = \frac{(t+2)^2}{1+t^2}$$

- Those who approached RHS this way gets 1/3 MAX from for the entire question as you've essentially done the prove up to * without t formula.

$$b) \text{ LHS} = \frac{(n+2)!}{[(n+2)-3]! 3!} - \frac{n!}{(n-3)! 3!}$$

$$= \frac{(n+2)!}{(n-1)! 3!} - \frac{n!}{(n-3)! 3!} \quad \text{--- (1)}$$

$$= \frac{n(n+1)(n+2)}{3!} - \frac{n(n-1)(n-2)}{3!}$$

$$= \frac{n[(n^2+3n+2) - (n^2-3n+2)]}{6}$$

$$= \frac{n \times 6n}{6} \quad \text{--- (1)}$$

$$= n^2$$

$$c) p=0.09 \quad q=0.91, \quad n=820$$

$$\sigma = \sqrt{npq} \doteq 8.195 \quad \text{--- (1)}$$

using formula for S.D

$$3\% \text{ of } 820 = 24.6$$

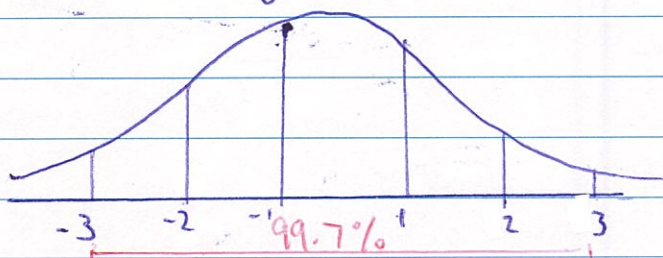
$$9\% \text{ of } 820 = 73.8$$

$\therefore$ looking for $P(49.2 \leq X \leq 98.4)$

$$Z_{49.2} = \frac{49.2 - 73.8}{\sigma} \doteq -3.002 \quad \text{--- (1)}$$

one of the Z-scores

$$Z_{98.4} = \frac{98.4 - 73.8}{\sigma} \doteq 3.002$$



$$\therefore P(|X - 73.8| > 24.6) = 0.03\% \quad \text{--- (1)}$$

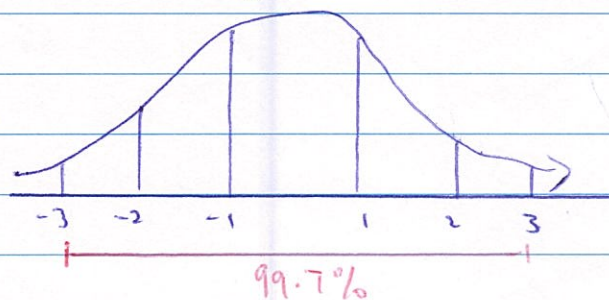
$$\sigma = \sqrt{\frac{pq}{n}} \doteq 9.994 \times 10^{-3} \quad \text{--- (1)}$$

using formula for S.D

$$P(0.06 \leq \hat{p} \leq 0.12) :$$

$$Z_{0.06} = \frac{0.06 - 0.09}{\sigma} \doteq -3.002 \quad \text{--- (1)}$$

$$Z_{0.12} = \frac{0.12 - 0.09}{\sigma} \doteq 3.002$$



$$\therefore P(|\hat{p} - 0.09| > 0.03) = 0.03\% \quad \text{--- (1)}$$

Q11 - marking standards

a) i) $h = 36 \sin(1.5t) - 15 \cos(1.5t) + 65$

→ ① $k = \sqrt{36^2 + 15^2} = \sqrt{1521} = 39$

either $\sqrt{1521}$ or 39 - 1 mark.

→ ② $\tan \alpha = \frac{15}{36} = \frac{5}{12} \Rightarrow \alpha = \tan^{-1}\left(\frac{5}{12}\right)$

or $\sin \alpha = \frac{5}{13}$

$\Rightarrow \alpha = \sin^{-1}\left(\frac{5}{13}\right)$

either

- 1 mark.

or $\cos \alpha = \frac{12}{13}$

$\Rightarrow \alpha = \cos^{-1}\left(\frac{12}{13}\right)$

α can have only one value.

or

$\alpha = 0.39479...$

(α can't be in degree form).

$= 22^\circ 37'$ or 22.62° (x)

→ ③ $39 \sin(1.5t - \tan^{-1}(\frac{5}{12}))$

expression

- 1 mark.

→ or any of the 4 angle forms above

ii) $h = 39 \sin(1.5t - \tan^{-1}(\frac{5}{12})) + 65 = 100$

no mark if sin is missing from the expression

→ ①

$39 \sin(1.5t - \tan^{-1}(\frac{5}{12})) = 35$

correct

sin ratio - 1 mark.

$\sin(1.5t - \tan^{-1}(\frac{5}{12})) = \frac{35}{39}$

radian form

degree form (ECF)

→ ② $1.5t - \tan^{-1}(\frac{5}{12}) = \sin^{-1}(\frac{35}{39})$

$1.5t = \tan^{-1}(\frac{5}{12}) + \sin^{-1}(\frac{35}{39})$

$\sin^{-1}(\frac{35}{39}) \doteq 63^\circ 49'$

$t = \frac{2}{3}(\tan^{-1}(\frac{5}{12}) + \sin^{-1}(\frac{35}{39}))$

$t = 57^\circ 37' 42.99''$

$= 57.6286^\circ$

correct t-value ① - 1 mark.

$= 1.0058089...$

→ or ③ $1.5t - \tan^{-1}(\frac{5}{12}) = \pi - \sin^{-1}(\frac{35}{39})$

$t = \frac{2}{3}(\pi - \sin^{-1}(\frac{35}{39}) + \tan^{-1}(\frac{5}{12}))$

$t = 92^\circ 31' 52.36''$

$= 92.53121^\circ$

correct t-value ② - 1 mark.

$= 1.614974...$

b) $(1 - 2x - 4x^2)(1 - \frac{3}{x})^9$

$$= (1 - 2x - 4x^2) ({}^9C_0 + {}^9C_1(-\frac{3}{x})^1 + {}^9C_2(\frac{-3}{x})^2 + \dots + {}^9C_5(\frac{-3}{x})^5 + {}^9C_6(\frac{-3}{x})^6 + {}^9C_7(\frac{-3}{x})^7$$

Terms in x^{-5} ${}^9C_5(-3)^5 - 2{}^9C_6(-3)^6 - 4{}^9C_7(-3)^7$ - 1 mark

$$= -30618 - 2 \times 61236 - 4 \times (-78732)$$

$$= -30618 - 122472 + 314928$$

$$= 161838$$

- 1 mark.

or 3 correct expressions but wrong calculation

- 1 mark/2

or $\begin{cases} 2 \text{ correct expressions} + \text{correct calculation} \\ 1 \text{ wrong expression} + \text{correct ECF calculation} \end{cases}$

- 1 mark/2

or in the expressions, forget to include "-" from "(-3)"

- 1 mark/2

or in the expressions, forget to consider "()" from "(-3)"

- 1 mark/2

$$\begin{aligned}
 \textcircled{12} \quad \frac{T_r}{T_{r+1}} &= \frac{\frac{50!}{r!(50-r)!} \times 0.15^r \times (0.85)^{50-r}}{\frac{50!}{(r+1)!(50-(r+1))!} \times 0.15^{r+1} \times 0.85^{50-(r+1)}} \\
 &= \frac{r+1}{50-r} \times \frac{0.85}{0.15} \\
 &= \frac{85(r+1)}{15(50-r)} \\
 &= \frac{17(r+1)}{3(50-r)}
 \end{aligned}$$

$$(ii) \quad P(x=r) \leq P(x=r+1)$$

$$\frac{P(x=r)}{P(x=r+1)} \leq 1 \quad \text{ie} \quad \frac{T_r}{T_{r+1}} \leq 1$$

$$\therefore \frac{17(r+1)}{3(50-r)} \leq 1$$

$$17r+17 \leq 150-3r$$

$$20r \leq 133$$

$$r \leq 6.65$$

$\therefore$ values will be $r=0, 1, 2, \dots, 6$.

(iii) most likely number of days late is 7.